超详细微分积分公式大全

一、微分公式全集

1. 基本函数导数

常数与幂函数

$$
\begin{aligned}
&\frac{d}{dx}c = 0\\
&\frac{d}{dx}x^n = nx^{n-1}\\
&\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}\\
&\frac{d}{dx}\sqrt[n]{x} = \frac{1}{n}x^{\frac{1}{n}-1}\\
&\frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}\\
&\frac{d}{dx}\frac{1}{x^n} = -\frac{n}{x^{n+1}}\\
&\frac{d}{dx}|x| = \operatorname{sgn}(x) = \begin{cases} 1, & x > 0\\ \text{未定义}, & x = 0\\ -1, & x < 0 \end{cases}
\end{aligned}
$$

指数函数

$$
\begin{aligned}
&\frac{d}{dx}e^x = e^x\\
&\frac{d}{dx}e^{kx} = ke^{kx}\\
&\frac{d}{dx}a^x = a^x \ln a \quad (a > 0, a \neq 1)\\
&\frac{d}{dx}a^{f(x)} = a^{f(x)} \ln a \cdot f'(x)\\
&\frac{d}{dx}e^{f(x)} = e^{f(x)} f'(x)
\end{aligned}
$$

对数函数

$$
\begin{aligned}
&\frac{d}{dx}\ln x = \frac{1}{x} \quad (x > 0)\\
&\frac{d}{dx}\ln |x| = \frac{1}{x} \quad (x \neq 0)\\
&\frac{d}{dx}\log_a x = \frac{1}{x \ln a} \quad (a > 0, a \neq 1)\\
&\frac{d}{dx}\ln f(x) = \frac{f'(x)}{f(x)}\\
&\frac{d}{dx}\log_a f(x) = \frac{f'(x)}{f(x) \ln a}
\end{aligned}
$$

2. 三角函数导数

基本三角函数

$$\begin{aligned}
&\frac{d}{dx}\sin x = \cos x \\
&\frac{d}{dx}\cos x = -\sin x \\
&\frac{d}{dx}\tan x = \sec^2 x = 1 + \tan^2 x \\
&\frac{d}{dx}\cot x = -\csc^2 x = -(1 + \cot^2 x) \\
&\frac{d}{dx}\sec x = \sec x \tan x \\
&\frac{d}{dx}\csc x = -\csc x \cot x
\end{aligned}$$

复合三角函数

$$\begin{aligned}
&\frac{d}{dx}\sin(ax+b) = a\cos(ax+b) \\
&\frac{d}{dx}\cos(ax+b) = -a\sin(ax+b) \\
&\frac{d}{dx}\tan(ax+b) = a\sec^2(ax+b) \\
&\frac{d}{dx}\sin f(x) = f'(x)\cos f(x) \\
&\frac{d}{dx}\cos f(x) = -f'(x)\sin f(x) \\
&\frac{d}{dx}\tan f(x) = f'(x)\sec^2 f(x)
\end{aligned}$$

3. 反三角函数导数

$$\begin{aligned}
&\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}} \quad (|x| < 1) \\
&\frac{d}{dx}\arccos x = -\frac{1}{\sqrt{1-x^2}} \quad (|x| < 1) \\
&\frac{d}{dx}\arctan x = \frac{1}{1+x^2} \\
&\frac{d}{dx}\operatorname{arccot} x = -\frac{1}{1+x^2} \\
&\frac{d}{dx}\operatorname{arcsec} x = \frac{1}{|x|\sqrt{x^2-1}} \quad (|x| > 1) \\
&\frac{d}{dx}\operatorname{arccsc} x = -\frac{1}{|x|\sqrt{x^2-1}} \quad (|x| > 1) \\
&\frac{d}{dx}\arcsin f(x) = \frac{f'(x)}{\sqrt{1-[f(x)]^2}} \\
&\frac{d}{dx}\arctan f(x) = \frac{f'(x)}{1+[f(x)]^2}
\end{aligned}$$

4. 双曲函数导数

$$\frac{d}{dx}\sinh x = \cosh x$$
$$\frac{d}{dx}\cosh x = \sinh x$$
$$\frac{d}{dx}\tanh x = \operatorname{sech}^2 x = 1 - \tanh^2 x$$
$$\frac{d}{dx}\coth x = -\operatorname{csch}^2 x = 1 - \coth^2 x$$
$$\frac{d}{dx}\operatorname{sech} x = -\operatorname{sech} x \tanh x$$
$$\frac{d}{dx}\operatorname{csch} x = -\operatorname{csch} x \coth x$$

5. 微分法则

线性性: $(cf \pm dg)' = cf' \pm dg'$

乘积法则: $(fg)' = f'g + fg'$

商法则: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

链式法则: $[f(g(x))]' = f'(g(x))g'(x)$

反函数法则: $(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$

二、积分公式全集

1. 基本不定积分公式

被积函数	原函数	条件/备注
x^n	$\frac{x^{n+1}}{n+1} + C$	$n \neq -1$
$\frac{1}{x}$	$\ln\|x\| + C$	$x \neq 0$
e^x	$e^x + C$	全体实数
e^{kx}	$\frac{1}{k}e^{kx} + C$	$k \neq 0$
a^x	$\frac{a^x}{\ln a} + C$	$a > 0, a \neq 1$
$\ln x$	$x\ln x - x + C$	$x > 0$
$\log_a x$	$\frac{x\ln x - x}{\ln a} + C$	$x > 0, a > 0, a \neq 1$

2. 三角函数积分

$$
\begin{aligned}
&\int \sin x\,dx = -\cos x + C \\
&\int \cos x\,dx = \sin x + C \\
&\int \tan x\,dx = -\ln|\cos x| + C = \ln|\sec x| + C \\
&\int \cot x\,dx = \ln|\sin x| + C = -\ln|\csc x| + C \\
&\int \sec x\,dx = \ln|\sec x + \tan x| + C \\
&\int \csc x\,dx = \ln|\csc x - \cot x| + C \\
&\int \sec^2 x\,dx = \tan x + C \\
&\int \csc^2 x\,dx = -\cot x + C \\
&\int \sec x \tan x\,dx = \sec x + C \\
&\int \csc x \cot x\,dx = -\csc x + C
\end{aligned}
$$

3. 三角幂次积分递推公式

$$
\begin{aligned}
&\int \sin^n x\,dx = -\frac{1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n}\int \sin^{n-2} x\,dx \\
&\int \cos^n x\,dx = \frac{1}{n}\cos^{n-1} x \sin x + \frac{n-1}{n}\int \cos^{n-2} x\,dx \\
&\int \tan^n x\,dx = \frac{1}{n-1}\tan^{n-1} x - \int \tan^{n-2} x\,dx \quad (n \neq 1) \\
&\int \cot^n x\,dx = -\frac{1}{n-1}\cot^{n-1} x - \int \cot^{n-2} x\,dx \quad (n \neq 1)
\end{aligned}
$$

4. 重要积分结果

$$
\begin{aligned}
&\int \sin^2 x\,dx = \frac{x}{2} - \frac{\sin 2x}{4} + C \\
&\int \cos^2 x\,dx = \frac{x}{2} + \frac{\sin 2x}{4} + C \\
&\int \sin^3 x\,dx = \frac{\cos^3 x}{3} - \cos x + C \\
&\int \cos^3 x\,dx = \sin x - \frac{\sin^3 x}{3} + C \\
&\int \sin^4 x\,dx = \frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C \\
&\int \cos^4 x\,dx = \frac{3x}{8} + \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C
\end{aligned}
$$

5. 反三角函数积分

$$
\begin{aligned}
\int \arcsin x\,dx &= x\arcsin x + \sqrt{1-x^2} + C \\
\int \arccos x\,dx &= x\arccos x - \sqrt{1-x^2} + C \\
\int \arctan x\,dx &= x\arctan x - \frac{1}{2}\ln(1+x^2) + C \\
\int \operatorname{arccot} x\,dx &= x\operatorname{arccot} x + \frac{1}{2}\ln(1+x^2) + C
\end{aligned}
$$

6. 含有平方项的积分

$$
\begin{aligned}
\int \frac{dx}{\sqrt{a^2-x^2}} &= \arcsin\frac{x}{a} + C \quad (|x|<|a|) \\
\int \frac{dx}{\sqrt{x^2+a^2}} &= \ln\left|x+\sqrt{x^2+a^2}\right| + C \\
\int \frac{dx}{\sqrt{x^2-a^2}} &= \ln\left|x+\sqrt{x^2-a^2}\right| + C \quad (|x|>|a|) \\
\int \frac{dx}{a^2+x^2} &= \frac{1}{a}\arctan\frac{x}{a} + C \\
\int \frac{dx}{x^2-a^2} &= \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C \quad (x\neq\pm a) \\
\int \frac{dx}{a^2-x^2} &= \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C \quad (|x|<|a|)
\end{aligned}
$$

7. 平方根积分

$$
\begin{aligned}
\int \sqrt{a^2-x^2}\,dx &= \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C \\
\int \sqrt{x^2+a^2}\,dx &= \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left|x+\sqrt{x^2+a^2}\right| + C \\
\int \sqrt{x^2-a^2}\,dx &= \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + C
\end{aligned}
$$

8. 双曲函数积分

$$
\begin{aligned}
\int \sinh x\,dx &= \cosh x + C \\
\int \cosh x\,dx &= \sinh x + C \\
\int \tanh x\,dx &= \ln(\cosh x) + C \\
\int \coth x\,dx &= \ln|\sinh x| + C \\
\int \operatorname{sech} x\,dx &= 2\arctan(e^x) + C \\
\int \operatorname{csch} x\,dx &= \ln\left|\tanh\frac{x}{2}\right| + C
\end{aligned}
$$

9. 分部积分常用结果

$$\int xe^{ax}\,dx = \frac{e^{ax}}{a^2}(ax-1)+C$$
$$\int x^2e^{ax}\,dx = \frac{e^{ax}}{a^3}(a^2x^2-2ax+2)+C$$
$$\int x\sin(ax)\,dx = \frac{1}{a^2}\sin(ax)-\frac{x}{a}\cos(ax)+C$$
$$\int x\cos(ax)\,dx = \frac{1}{a^2}\cos(ax)+\frac{x}{a}\sin(ax)+C$$
$$\int e^{ax}\sin(bx)\,dx = \frac{e^{ax}(a\sin bx-b\cos bx)}{a^2+b^2}+C$$
$$\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}(a\cos bx+b\sin bx)}{a^2+b^2}+C$$
$$\int \ln x\,dx = x\ln x - x + C$$
$$\int x\ln x\,dx = \frac{x^2}{2}\ln x-\frac{x^2}{4}+C$$

10. 有理函数积分（部分分式法）

$$\int \frac{dx}{x-a} = \ln|x-a|+C$$
$$\int \frac{dx}{(x-a)^n} = \frac{-1}{(n-1)(x-a)^{n-1}}+C \quad (n>1)$$
$$\int \frac{dx}{x^2+px+q} = \frac{2}{\sqrt{4q-p^2}}\arctan\frac{2x+p}{\sqrt{4q-p^2}}+C \quad (p^2<4q)$$
$$\int \frac{x\,dx}{x^2+px+q} = \frac{1}{2}\ln|x^2+px+q|-\frac{p}{2}\int\frac{dx}{x^2+px+q}$$
$$\int \frac{dx}{(x-a)(x-b)} = \frac{1}{a-b}\ln\left|\frac{x-a}{x-b}\right|+C \quad (a\neq b)$$

11. 定积分重要公式

$$\int_0^\infty e^{-ax}\,dx = \frac{1}{a}, \quad a>0$$
$$\int_0^\infty x^ne^{-ax}\,dx = \frac{n!}{a^{n+1}}, \quad a>0, n\in\mathbb{N}$$
$$\int_{-\infty}^\infty e^{-x^2}\,dx = \sqrt{\pi}$$
$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$$
$$\int_0^\infty \frac{1-\cos x}{x^2}\,dx = \frac{\pi}{2}$$
$$\int_0^{\pi/2}\sin^n x\,dx = \int_0^{\pi/2}\cos^n x\,dx = \begin{cases}\dfrac{(n-1)!!}{n!!}\cdot\dfrac{\pi}{2}, & n\text{偶数}\\ \dfrac{(n-1)!!}{n!!}, & n\text{奇数}\end{cases}$$
$$\int_0^1 x^{m-1}(1-x)^{n-1}\,dx = B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} \quad (\text{Beta函数})$$

三、积分技巧与方法

1. 换元积分法

$$
\begin{aligned}
&\text{第一类换元（凑微分）：} \int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x) \\
&\text{第二类换元:} \\
&\quad \sqrt{a^2 - x^2} : \text{令 } x = a\sin\theta,\ dx = a\cos\theta d\theta \\
&\quad \sqrt{a^2 + x^2} : \text{令 } x = a\tan\theta,\ dx = a\sec^2\theta d\theta \\
&\quad \sqrt{x^2 - a^2} : \text{令 } x = a\sec\theta,\ dx = a\sec\theta\tan\theta d\theta \\
&\text{倒代换: } x = \frac{1}{t},\ dx = -\frac{1}{t^2}dt
\end{aligned}
$$

2. 分部积分法

$$\int u\,dv = uv - \int v\,du$$

常用配对：

- 幂函数×指数函数：$u = x^n$, $dv = e^{ax}dx$
- 幂函数×三角函数：$u = x^n$, $dv = \sin(ax)dx$ 或 $dv = \cos(ax)dx$
- 指数函数×三角函数：$u = e^{ax}$, $dv = \sin(bx)dx$ 或 $dv = \cos(bx)dx$
- 对数函数×幂函数：$u = \ln x$, $dv = x^n dx$
- 反三角函数×幂函数：$u = \arcsin x$, $dv = dx$

3. 有理函数积分步骤

1. 如果分子次数⩾分母次数，多项式除法 2. 分母因式分解 3. 部分分式分解 4. 逐项积分

4. 三角函数积分技巧

- 奇次幂：提取一个因子，用 $\sin^2 x + \cos^2 x = 1$
- 偶次幂：用倍角公式降幂
- 乘积形式：用积化和差公式
- 万能代换：$t = \tan\frac{x}{2}$, $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$

5. Wallis公式（三角幂次定积分）

$$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \begin{cases} \frac{\pi}{2}\cdot\frac{(n-1)!!}{n!!}, & n\text{为偶数} \\ \frac{(n-1)!!}{n!!}, & n\text{为奇数} \end{cases}$$

其中 $n!!$ 表示双阶乘。

四、特殊函数积分

Gamma函数

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \quad \Re(z) > 0$$

性质：

$$\begin{aligned}
&\Gamma(1) = 1, \quad \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \\
&\Gamma(z+1) = z\Gamma(z) \\
&\Gamma(n) = (n-1)! \quad (n \in \mathbb{N}^+)
\end{aligned}$$

Beta函数

$$B(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\, dt = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}, \quad p,q > 0$$

误差函数

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

性质：$\operatorname{erf}(0) = 0$, $\operatorname{erf}(\infty) = 1$

五、常用积分表（速查）

被积函数	积分结果
$\int \frac{dx}{x}$	$\ln\lvert x\rvert + C$
$\int \frac{dx}{x^2+a^2}$	$\frac{1}{a}\arctan\frac{x}{a} + C$
$\int \frac{dx}{\sqrt{a^2-x^2}}$	$\arcsin\frac{x}{a} + C$
$\int \frac{dx}{\sqrt{x^2\pm a^2}}$	$\ln\lvert x+\sqrt{x^2\pm a^2}\rvert + C$
$\int \sqrt{a^2-x^2}\, dx$	$\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$
$\int \sqrt{x^2\pm a^2}\, dx$	$\frac{x}{2}\sqrt{x^2\pm a^2} \pm \frac{a^2}{2}\ln\lvert x+\sqrt{x^2\pm a^2}\rvert + C$
$\int e^{ax}\cos(bx)\, dx$	$\frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C$
$\int e^{ax}\sin(bx)\, dx$	$\frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C$
$\int \ln x\, dx$	$x\ln x - x + C$
$\int \sin^2 x\, dx$	$\frac{x}{2} - \frac{\sin 2x}{4} + C$
$\int \cos^2 x\, dx$	$\frac{x}{2} + \frac{\sin 2x}{4} + C$